One continuation-connected family and two bracketed Floquet events in the Li–Li–Liao unequal-mass catalog

Yossi Eliaz
Incredibuild, incredibuild.com
islo.dev and
Department of Computer Science, Holon Institute of Technology (HIT), Holon, Israel

Ori Chemo
Incredibuild, incredibuild.com
(Dated: August 19, 2026)

The Li–Li–Liao catalog reports 135,445 unequal-mass non-hierarchical planar periodic orbits of syzygy word *bABbabaBAba* on a 0.001 mass grid; a later analysis of the same table reads it as two families. Treating family identity as continuation connectivity, and the stability boundary as the preimage of a universal spectral domain rather than a stable/unstable mask, we find the sampled set supported as one continuation-connected component on 26 adversarially chosen links, and the scale-invariant period/angular-momentum projection on this sheet folded and non-injective—so two visible branches do not imply two families. Two transitions are independently reproduced in a second arbitrary-precision program as bracketed mechanisms, a physical $+1$ crossing and an opposite-Krein Hamiltonian–Hopf collision, to m_2 widths 1.578×10^{-9} and 6.268×10^{-9}; three mixed roots organizing the $+1/-1$ switches are supported at 60 digits but localized by residual, one rung lower. We further report a limit of the method: the frozen 2×10^{-8} event gate is finer than the reproducibility of the quantity it gates, a rotation re-derivation at identical tolerance moving the event on 345 of 620 transition cells by more than the gate itself. The critical graph is consequently not closed and the status is open. "Stable" means planar linear Floquet stability only.

Introduction.—The general three-body problem is not solved here, and nothing below bears on it. The object is finite: one frozen catalog and its stability geometry. Li, Li and Liao sampled 135,445 unequal-mass non-hierarchical planar periodic orbits on a 0.001 grid over $m_1 \in [0.8, 1.1]$, $m_2 \in [0.7, 1.2]$ at $m_3 = 1$, labelled 13,315 linearly stable, and left 620 adjacent cells across which that label flips [1]. The orbits share the syzygy word *bABbabaBAba* that the source literature attaches to this catalog. Removing the three binary-collision points from the shape sphere leaves a space whose fundamental group is free on two generators, so a collisionless periodic orbit has a free homotopy class, that is a conjugacy class in F_2: a cyclic word modulo free reduction and rotation, with an uppercase letter denoting the inverse generator. We carry the word as cited catalog metadata and derive nothing from it. It is already freely and cyclically reduced, canonically *ABbabaBAbab*. We do not assert a letter-per-syzygy correspondence: letter count and collinear-crossing count are distinct, and the number of syzygies per period on this sheet is not a quantity this work has verified. The word is a topological label, not a dynamical one, and we never use it as family identity: distinct branches may share a free-group conjugacy class.

Two questions about that frozen object are answerable rather than philosophical. Is the sampled set one family, where *family* means branch-preserving continuation inside the box rather than the appearance of a projection—the distinction that separates this reading from the two-set reading of the same table [2]? And what is the stability boundary, resolved by mechanism, once one stops treating stability as a binary mask?

Spectral frame.—For the centre-of-mass-reduced planar problem the reduced monodromy is symplectic on the physical E^{ω}/E quotient and its spectrum is fixed by two invariants (α, β). Spectral stability is then a region $\mathcal{D}$ in that plane bounded by the walls $G_+ = 0$ and $G_- = 0$, where an eigenvalue reaches $+1$ or -1, and by the discriminant parabola $\Delta = 0$, where two unit-circle pairs collide; the three curves meet at the exact vertices $(0, -4)$, $(4, 4)$ and $(8, 28)$. This algebra is classical symplectic Floquet and Krein theory [3, 4] and is not a result of this work. What is computed is the preimage $\Phi^{-1}(\partial\mathcal{D})$ on one catalog sheet, under $\Phi : (m_1, m_2) \mapsto (\alpha, \beta)$. The gain is that a transition acquires a mechanism—a G_+ crossing, a G_- crossing, or a Δ collision—with the $(4, 4)$ vertex as the organizing preimage where a G_+ and a G_- wall meet. Figure 1 shows both planes.

Claims below carry the rung of the artifact that produced them. *Screening* is binary64 shooting and event localization at two gates never loosened in this work, maximum absolute event 2×10^{-8} and maximum periodic closure 10^{-7}; a binary64 point is a proposal. *Supported* means recomputed at 60 digits in an independent program with canonical Jacobi monodromy and the physical symplectic quotient, but localized by Newton residual. *Independently reproduced* means localized by a sign change of the event function in a program sharing no dynamics, shooting or Floquet implementation with the screening stack. The separation between a sign change and a residual is not stylistic, and we keep it visible.

One component.—Within the frozen catalog and the

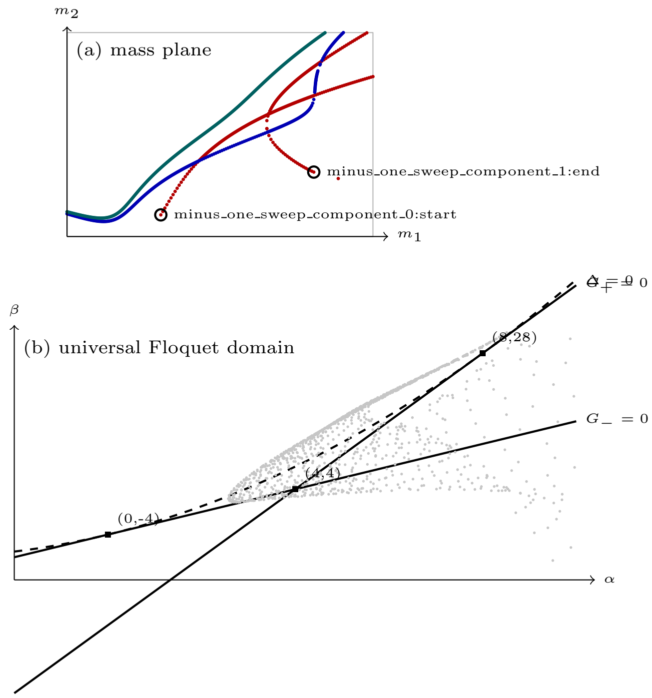

FIG. 1. (a) All 775 localized critical roots in the declared mass rectangle, by mechanism: G_+ crossings, G_- crossings and Δ collisions. Open circles are the two polyline ends the assembler leaves unattached. (b) The universal Floquet domain in the smooth invariants, with the exact walls $G_+ = \beta - 6\alpha + 20$, $G_- = \beta - 2\alpha + 4$ and $\Delta = (\alpha - 4)^2 - 4(\beta - 4\alpha + 8)$, their three exact vertices, and the sheet's converged probe images (every 24th of 30916, thinned only for legibility). Every mark is a committed measurement; nothing is fitted or interpolated.

declared normalization the sampled set is supported as one continuation-connected component. The evidence is adversarial rather than exhaustive. On a 135,444-edge spanning tree of the 270,087 mass-adjacent links we selected 26 links to be as hostile to connectivity as the data permits: five macroscopic bottleneck cuts, the twenty globally worst chart jumps, and one deliberately pathological invariant-near-duplicate bridge whose endpoints lie 0.0617 apart in mass—nearly sixty-two grid steps—and which is therefore not a tree edge at all. Each was crossed bidirectionally without loosening a gate. Chart independence was demonstrated on exactly one of the 26, the pathological bridge, by re-crossing it in a generic strict-periodic chart.

The two-family reading is addressed on the mechanism rather than on the labels. On this sheet the corrected scale-invariant (T_{si}, L_{si}) projection is folded and non-injective: 590 mass-adjacent edges carry a determinant sign change and assemble into one fold locus, and 86 far-separated mass pairs agree to standardized invariant distance below 10^{-4}. A folded non-injective projection can show two visible branches for one connected sheet, so the inference from two branches to two dynamical families does not follow. This does not disprove the earlier analysis on its own terms [2]. Three caveats are not negotiable: the certificate covers 26 links and not 135,444, chart independence rests on one of them, and separation below the published 0.001 grid is not excluded.

TABLE I. The five objects, each at the rung its own artifact declares.

Object	Location	Rung
+1 crossing	$m_1 = 0.80234461$, $\delta m_2 = 1.578 \times 10^{-9}$	reproduced
H–Hopf collision	$m_1 = 0.80228898$, $\delta m_2 = 6.268 \times 10^{-9}$	reproduced
Mixed root I	$(0.92923919, 0.88536646)$	supported
Mixed root II	$(0.99676820, 0.95601936)$	supported
Mixed root III	$(1.04955319, 1.12947581)$	supported

Two bracketed events, three mixed roots.—On the slice $m_1 = 0.8023446113666945$ the lower transition is a generic physical +1 crossing of the reduced return map, bracketed in m_2 to width $1.5776460402494633 \times 10^{-9}$. On $m_1 = 0.8022889780964406$ the upper transition is an opposite-Krein Hamiltonian–Hopf collision, bracketed to $6.2677772144967 \times 10^{-9}$, with two upper-half unit modes of opposite Krein sign on the stable flank and none on the unstable flank; the Krein signature, not merely the eigenvalue positions, identifies the mechanism. One rung below, the three mixed roots of $G_+ = G_- = 0$—preimages of the $(4, 4)$ vertex—are recomputed at 60 digits with relative event norms from 1.2×10^{-14} to 1.5×10^{-13}, but each is localized by Newton residual rather than by a sign-change bracket. The same five objects, relocalized at 50 and 60 digits and re-derived through a canonical-Jacobi and physical-quotient layer, agree in mass to between 8.9×10^{-27} and 5.7×10^{-25}; those two arbitrary-precision paths share their bracket-localization code, so this is precision invariance and not a second program. A 60-digit bracket is a bracket, not an interval-arithmetic existence proof; a residual is weaker still. That distinction is not rhetorical here. A rigorous validated-flow enclosure was attempted at five seeds with a C^1 variational interval integrator; four fail outright, and the one returning a finite enclosure that excludes collision encloses the Floquet invariants only to $\alpha \in [-1.4, 1.4] \times 10^{12}$ and $\beta \in [-9.4 \times 10^{23}, 1.9 \times 10^{24}]$, with a discriminant interval straddling zero by $\pm 7.5 \times 10^{24}$ and a closure interval 3.1×10^{-3} wide against the 10^{-7} gate — enclosures admitting every stability classification at once, and whose own record calls itself scaffolding and declines to be an existence proof.

The 620 cells.—All 620 published transition cells are localized once each, none left as a Newton failure, and each is assigned to exactly one of seven mechanism polylines: 198 on G_+ crossings, 168 on G_-, and 254 on Δ collisions. Six further label-invisible event-sign polylines from a full-domain sweep carry 153 cells, giving a thirteen-edge assembly. That assembly is a mixture and not a certification: by the edges' own estimator records, seven of the thirteen polylines rest on binary64 screening alone and six carry independent arbitrary-precision corroboration, and of the 620 cells 462 are binary64 and 158 arbitrary-precision. Under the ladder above, more than half the decomposition is a proposal.

A limit of the method.—The frozen event gate is finer than the reproducibility of the quantity it gates, and we report this rather than the headline alone. Across the 775 localized roots the worst absolute event is $1.989798081858396 \times 10^{-8}$ against the 2×10^{-8} gate—99.49% of the budget, cleared by 0.51%, with 222 roots above 10^{-8}. That number is a property of one chart, one tolerance and one machine rather than of the roots: re-deriving the same monodromy through a rotation at identical tolerance moves the event by more than the gate on 345 of the 620 cells and by more than ten times the gate on 144, concentrated in G_+ (126/198) and Δ (217/254), with G_- (2/168) the only mechanism whose compliance survives. Conditioning is recorded for none of the 775 roots. Closure certificates are optimistic for a related reason: they are written from the corrector's screening integration at relative tolerance 2×10^{-10} while the event is evaluated at 5×10^{-13}, and on a seeded sample of 160 roots the recorded closure is optimistic by a median factor 48.6, four exceeding the 10^{-7} closure gate on re-measurement. Mechanism labels are not immune: arbitrary-precision escalation of sixteen binary64-failed cells returned four as Δ collisions where binary64 had recorded G_+. The same gap appears as displacement rather than as chart dependence. At the three tip masses binary64 screening returned events $+1.53\times10^{-8}$, -1.02×10^{-8} and -3.82×10^{-10}, all inside the gate; re-evaluated at 60 digits at those same masses the events are -1.116×10^{-6}, -1.131×10^{-6} and $+4.298 \times 10^{-8}$, exceeding it by factors 55.8, 56.6 and 2.1, and the bracketed roots lie 7.8×10^{-7}, 5.1×10^{-7} and 5.2×10^{-7} away in m_2. A binary64 event inside this gate is therefore not evidence that a root sits where binary64 placed it.

What is not established.—The critical graph is not closed. Of 26 edge endpoints 24 are attached; two finite-lattice sweep ends remain unclassified, at $(0.892, 0.75308)$ and $(1.042, 0.85798)$. The incidence splits into two components—the collision curve is still its own—and three declared nodes carry no incidence. Both unclassified ends sit above their local non-existence frontier, not inside it: orbits still close at the ends themselves, with re-corrected closures 6.6×10^{-8} and 3.9×10^{-8} inside the 10^{-7} gate. What defeats them is that continuing outward drives the event functional non-finite after travelling only 1.4×10^{-3} and 2.2×10^{-3} in m_2. The tips themselves are bracketed roots rather than proposals: at 60 digits the event function changes sign across m_2 intervals of width 3.354856×10^{-9}, 3.877628×10^{-9} and 3.865492×10^{-9} about $(0.892, 0.7530796)$, $(1.042, 0.8579752)$ and $(1.046, 1.1060303)$, with bracket-endpoint closures at or below 6.2×10^{-21}. What is unclassified at the two ends is therefore the continuation past them and not their existence; no mechanism is claimed for the three, their artifacts declining that rung. Neither is therefore a catalog overlap, a declared-domain face, nor a closed loop, which are the only stops the continuation resolver admits; we have implemented a fourth, an existence-boundary terminus, whose award requires divergence rather than a marginal miss, invariance of the failure under step refinement and under two independent higher-precision paths, independent audit corroboration, and outward travel. Every one of those conditions can only refuse, and on the present evidence the class is awarded to neither end, so the two remain unclassified here. Three further classifications are explicitly not claimed. The secondary-left birth is not a certified nondegenerate fold: transversality is real, with $\partial G/\partial m_1 = 28.0456$ convergent under refinement, and stationarity is met, $\partial G/\partial m_2$ falling to -6.1×10^{-14}; what is missing is the curvature, since $\partial^2 G/\partial m_2^2$ does not converge under stencil refinement but grows as h^{-2}, from -3.9×10^8 at $h = 8 \times 10^{-5}$ to -2.5×10^{10} at $h = 10^{-5}$, so the quadratic normal form is not established. The secondary-right terminus carries physical evidence only and is typed an endpoint, so no fourth organizer is claimed. The lower $+1$ daughter is not classified, since the classifier emits a single outcome by construction and its recorded class is therefore an assertion rather than a discrimination.

A coverage condition that ranges over non-existence.—One release condition asks that a sign-vector audit leave no gap wider than the 10^{-3} mass grid step between consecutively converged probes, across the whole declared span $m_2 \in [0.7, 1.2]$, on every scan line. It is worth reporting why that condition cannot be met, because the reason is not sampling density. In the 465-probe audit the condition reads, 100 probes fail, and their closure residuals have median 1.56 against a 10^{-7} gate — seven orders outside it, with 97% of failures above 10^{-2}. In the densest audit available, 33,701 probes, 2,785 fail with median closure 3.15 and again 97% above 10^{-2}. The failures are not marginal: no periodic orbit of this family closes at those masses at all. They concentrate where one expects, below the family's lower reach — 2,660 of 2,785 at $m_2 < 0.9$, failed median m_2 0.739 against 0.969 for converged probes. A condition demanding converged coverage of the full declared rectangle therefore ranges over a region in which the object to be covered does not exist, and no additional computation can satisfy it. The honest form of the statement is scoped to the existence support of the family, and the frontier itself — traced on more than thirty scan lines, and rising monotonically in m_1 only above $m_1 \approx 0.995$ — is a first-class object this work has located but not yet certified.

Completeness, bounded.—Completeness is frozen and bounded, and the bound is small. As a two-dimensional region the frozen neck raster covers $m_1 \in [0.997, 0.999] \times m_2 \in [0.993, 1.012]$ at step 10^{-4} with 4011 samples, all 21 lines separated: 0.0253% of the declared area. The only other input is a twelve-proposal off-grid screen in one pocket, 0.002539%, whose own record describes it as machine-proposed candidates with binary64 screening and not discovery evidence. The two together are 0.0279% of the box; everything else is probed on a measure-zero union of scan lines. So completeness here means only that no additional critical object was detected by a finite sign-change sampling at those resolutions. Even-order root pairs and tangencies are explicitly not excluded, nor are sub-resolution loops or curves between scan lines. Whether the interior is clean is unresolved: the full-domain sweep reports gate-passing roots matching no committed edge.

Status.—Two release conditions are false: no unclassified edge endpoints, and sign-topology coverage. The upstream table is rejected unless its content digest matches, environments are pinned, and the release bit is writable only by the assembler, which reproduces the committed critical-graph artifact byte for byte and exits nonzero while the problem is open. The scientific status is therefore open, this Letter reports what is earned rather than a solution, and no priority claim is made. Throughout, "stable" denotes spectral stability of the linearized return map for planar motion—not nonlinear or Lyapunov stability, not KAM, and not stability against non-planar perturbation. A spectrally stable Hamiltonian orbit can be nonlinearly unstable through resonance.

Data and code availability.—Every number above is read from a committed artifact in the repository that produced it, and each is mapped to its artifact and evidence rung in the accompanying claim ledger. The upstream catalog is not redistributed: it is fetched at run time and rejected unless its Git blob hash is exactly `79b2963df43e62201c35690bfc22bec166132427`. Python and Julia environments are pinned by lockfile and manifest, the release bit is writable only by the assembler, and the assembler reproduces the committed critical-graph artifact byte for byte. Figure 1 is generated from those artifacts by a standard-library script in the repository, so it carries no plotting dependency and its every mark is text-diffable.

The analysis pipeline, the claim-gating machinery and this manuscript were produced by AI coding agents operating in islo.dev sandboxes under the authors' direction. No agent output is admitted as evidence above the *candidate* rung, and no claim is stated above the rung its own artifact declares: the two bracketed events and the three mixed roots were relocalized by the arbitrary-precision stack described above, while what rests on binary64 screening alone—462 of the 620 cells, seven of the thirteen polylines, and the audits of the preceding section—is labelled as such in the record. Numerical verification ran in GitHub Actions and in ephemeral cloud instances.

[1] X. Li, X. Li, and S. Liao, One family of 13315 stable periodic orbits of the non-hierarchical unequal-mass triple system, Science China Physics, Mechanics & Astronomy **64**, 219511 (2021), arXiv:2007.10184 [nlin.CD].

[2] D. Stančević, V. Vasiljević, and V. Dmitrašinović, The mass-, period- and angular momentum dependences of the "moth-i" family of 3-body orbits, SSRN preprint (2023).

[3] V. A. Yakubovich and V. M. Starzhinskii, *Linear Differential Equations with Periodic Coefficients* (Wiley, New York, 1975) translated from the Russian; standard reference for the strong-stability and Krein-signature theory of symplectic monodromy used here as a frame.

[4] U. Frauenfelder, D. Koh, and A. Moreno, Symplectic methods in the numerical search of orbits in real-life planetary systems, SIAM Journal on Applied Dynamical Systems **22**, 3284 (2023).